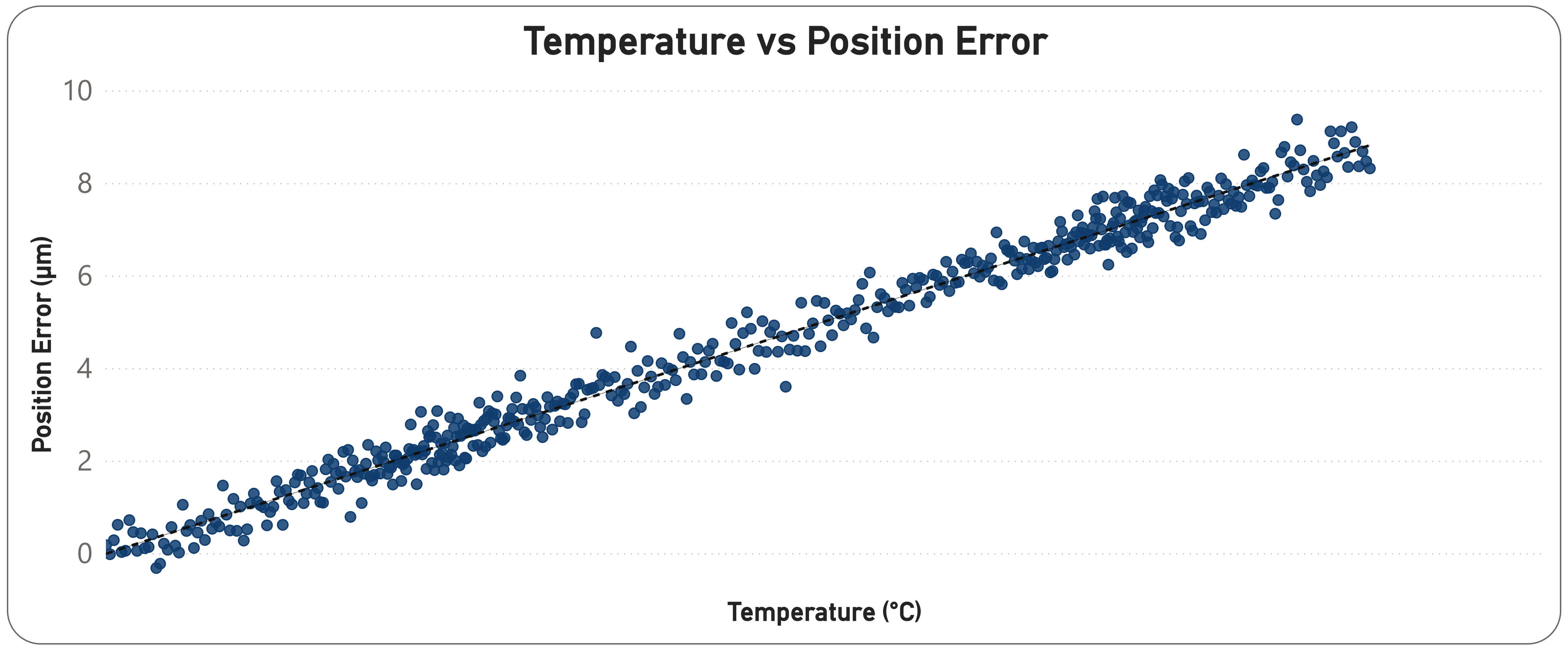
# Thermal Drift & Compensation Analysis in Precision Machine System

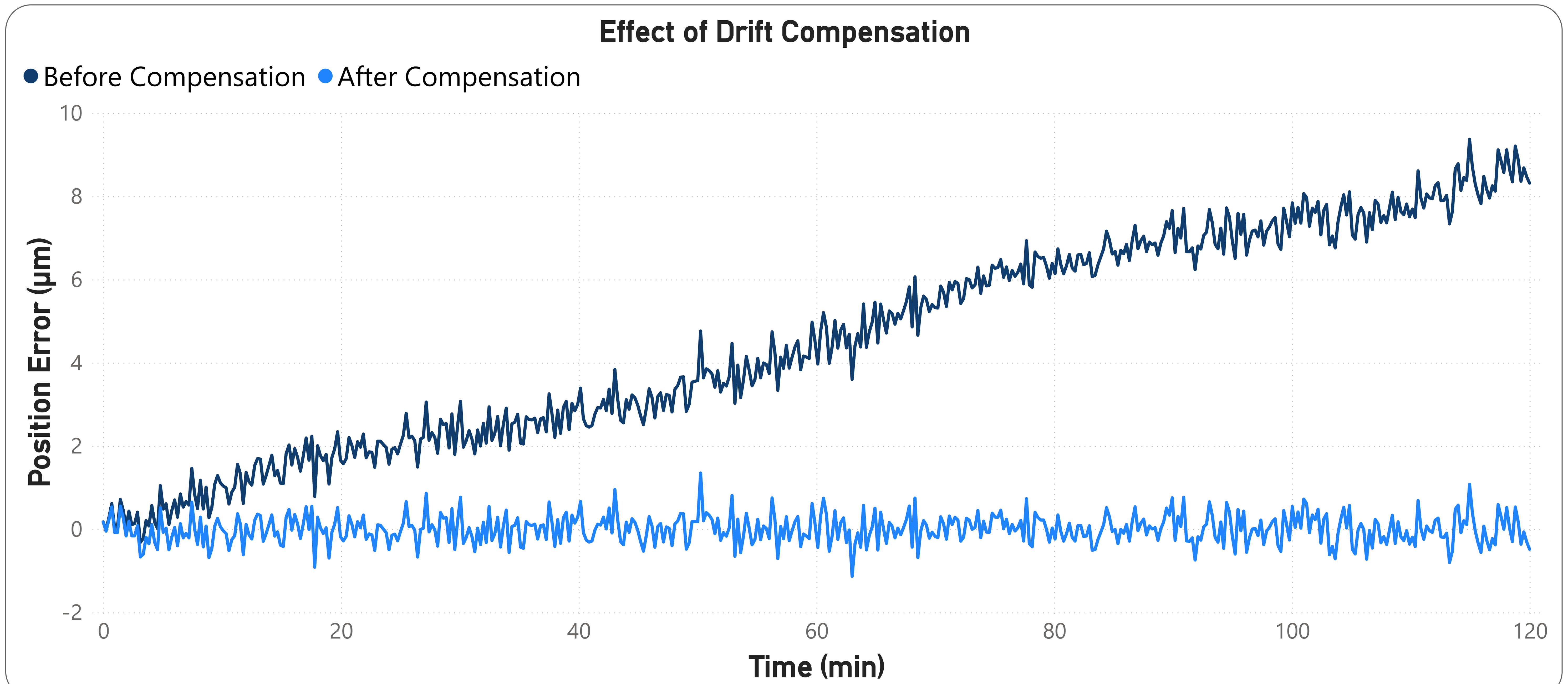
*A Visual Engineering Study: Simulation & Analysis of Temperature Induced Position Drift*

**Thermal drift behavior during machine warm up & evaluation of temperature based compensation**

## *Problem & Objective*

Thermal expansion during machine warm up can introduce positioning errors at micron level in precision machine axes. This project analyzes the relationship between temperature rise & positioning drift using simulated data and visual analysis, & evaluates a compensation strategy based on temperature to reduce drift & improve positioning accuracy.




## *Model*

***e(t) = k · (T(t) − T_ref) + n(t)***

- e(t): measured position error
- k: thermal drift sensitivity
- T(t): temperature
- T_ref: reference temperature
- n(t): measurement noise

**Compensation model**

***e_comp(t) = e(t) − k (T(t) − T_ref)***

| | | | |
|---|---|---|---|
| 3.52 | 9.37 | 1.35 | 85.61% |
| **Temperature Rise** | **Maximum Drift (μm)** | **Residual Error (μm)** | **Improvement (%)** |

***Thermal drift during machine warm up produces a maximum positioning error of 9.37 μm. Applying a temperature based compensation model reduces the residual error to 1.35 μm, improving positioning accuracy by 85.6 %.***